NEERAJ KUMAR PANDIT

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EDUCATION

University of Göttingen Göttingen, DE
Ph.D. in Chemistry (Thesis Advisor: Prof. Lutz Ackermann) Aug 2022 – *present*

- Research funded by SPP2363 Molecular Machine Learning (DFG, Germany)
- Thesis: Modeling and mechanistic studies on asymmetric C–H activation

Indian Institute of Technology Indore Indore, IN
M.Sc. in Chemistry (Thesis Advisor: Prof. Biswarup Pathak) Aug 2020 – May 2022

- Thesis: Rational Designing of Bimetallic/Trimetallic HER Catalysts through Supervised Machine Learning
- CGPA: 8.76/10

University of Delhi, Rajdhani College New Delhi, IN
B.Sc. Hons. in Chemistry Aug 2017 – May 2020

- CGPA: 7.703/10

EXPERIENCE

University of Göttingen Göttingen, DE
Doctoral Researcher – Computational Study of Asymmetric Catalysis Aug 2022 – present

- Used DFT-based feature analysis and regression models to design and evaluate new chiral ligands for nickel-catalyzed asymmetric transformations.
- Analyzed structure–activity relationships of chiral ligands and applied classification models to explore reactivity thresholds in cobalt-catalyzed systems.

Indian Institute of Technology Indore Indore, IN
Master's Researcher – Computational Screening of HER Electrode Materials Aug 2021 – May 2022

- Screened over 5000 candidate materials for hydrogen evolution catalysis using DFT (GPAW) and supervised machine learning trained on a small, curated dataset.

SKILLS

Programming & Tools: Python, Linux (CLI), LaTeX, HTML
Libraries & Frameworks: NumPy, SciPy, Pandas, Scikit-learn, RDKit, PyTorch
Computational Chemistry: Gaussian, ORCA, xTB, GPAW, CPPAW, ASE
Techniques: DFT, ligand design, structure-activity relationships, feature extraction, compound library screening
Languages: English (fluent), German (elementary), Hindi (native)

PUBLICATIONS

§These authors contributed equally.

[1] Y. Xu, N. K. Pandit, S. Meraviglia, P. Boos, P. A. M. Stark, M. Surke, R. H.-Irmer, D. Stalke, L. Ackermann, "Stereocontrolled Construction of Multi-Chiral [2.2]Paracyclophanes via Cobaltaphotoredox Dual Catalysis", *ACS Catal.* **2025**, *15*, 11716–11725.

[2] P. Boos, N. K. Pandit, S. Dana, T. von Münchow, A. Hashidoko, S. Trienes, L. Haberstock, R. H.-Irmer, D. Stalke, L. Ackermann, "Multiple Atropo Selectivity by k2-N,O-Oxazoline Urea Ligands in Cobaltaelectro-Catalyzed C–H Activations: Decoding Selectivity with Data Science Integration", *ChemistryEurope* **2025**, *3*, e202500071.

[3] T. von Münchow,§ N. K. Pandit,§ S. Dana, P. Boos, S. E. Peters, J. Boucat, Y.-R. Liu, A. Scheremetjew, L. Ackermann, "Enantioselective C–H annulations enabled by either nickel- or cobalt-electrocatalysed C–H activation for catalyst-controlled chemodivergence ", *Nat. Catal.* **2025**, *8*, 257-269.

[4] S. Dana,§ N. K. Pandit,§ P. Boos,§ T. von Münchow, S. E. Peters, S. Trienes, L. Haberstock, R. H.-Irmer, D. Stalke, L. Ackermann, "Parametrization of k2-N,O-Oxazoline Preligands for Enantioselective Cobaltaelectro-catalyzed C–H Activations ", *ACS Catal.* **2025**, *15*, 4450-4459.

[5] S. Cattani, N. K. Pandit, M. Buccio, D. Balestri, L. Ackermann, G. Cera, "Iron-Catalyzed C–H Alkylation/Ring Opening with Vinylbenzofurans Enabled by Triazole", *Angew. Chem. Int. Ed.* **2024**, *63*, e202404319.

[6] T. Michiyuki, S. L. Homölle, N. K. Pandit, L. Ackermann, "Electrocatalytic Formal C(sp2)–H Alkylations via Nickel-Catalyzed Cross-Electrophile Coupling with Versatile Arylsulfonium Salt", *Angew. Chem. Int. Ed.* **2024**, *63*, e202401198.

[7] K. Zeng, N. K. Pandit, J. C. A. Oliveira, S. Dechert, L. Ackermann, K. Zhang, "Weak-coordination-auxiliary aminocatalysis enables directed [3 + 2] cyclization for 2-acylindolizines", *Green Chem.* **2024**, *26*, 5253-5259.

[8] N. K. Pandit, D. Roy, S. C. Mandal, B. Pathak, "Rational Designing of Bimetallic/Trimetallic Hydrogen Evolution Reaction Catalysts Using Supervised Machine Learning", *J. Phys. Chem. Lett.* **2022**, *13*, 7583-7593.

PRESENTATIONS

- N. K. Pandit, T. v. Münchow, S. Dana, P. Boos, S. E. Peters, J. Boucat, Y.-R. Liu, A. Scheremetjew, S. Trienes, L. Haberstock, R. H.-Irmer, D. Stalke, L. Ackermann, "Data-Driven Strategies and Mechanistic Insights into Electrochemical Ni/Co-Catalyzed Enantioselective C–H Activation", Presented at *17th Göttinger ChemieForum*, Göttingen, Germany, Jun 2025. - Poster presentation
- N. K. Pandit, P. Boos, L. Ackermann, "Nickel-electrocatalyzed enantioselective C–H activations for chemo-divergence", Presented at *International Leopoldina Symposium on Molecular Machine Learning*, Halle (Saale), Germany, Mar 2024. - Poster presentation
- N. K. Pandit, P. Boos, L. Ackermann, "Machine Learning for Developing and Understanding, Asymmetric 3d Metal-Catalyzed C–H Activations", Presented at *Dream Reactions with (and without) Hydrogen*, Münster, Germany, Aug 2023. - Poster presentation
- N. K. Pandit, P. Boos, L. Ackermann, "Machine Learning for Developing and Understanding, Asymmetric 3d Metal-Catalyzed C–H Activations", Presented at *SPP2363 kickoff meeting*, Halle (Saale), Germany, Oct 2022. - Oral and poster presentation

WORKSHOPS AND COURSES

International CP-PAW Autumn School — University of Göttingen
Hands-on Course on Density-Functional Calculations — Sep 2024
- Performed DFT calculations using CPPAW.

Fundamentals of Deep Learning — NVIDIA DLI
Instructor-Led Workshop — Feb 2022
- Fundamental techniques and tools required to train a deep learning model, common deep learning data types and model architectures, and transfer learning.

Qubit by Qubit — The Coding School
Introduction to Quantum Computing Course — Oct 2020 – May 2021
- Quantum mechanics, quantum information and computation, quantum algorithms, Quiskit, and IBM Quantum Experience to run quantum simulations on real quantum hardware.